



TRANSFORMERS

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PREPROCESSING

- Used bert-base-uncased tokenizer
- Cleaned spacing
- No punctuation removal
- No stopword removal
- No stemming/lemmatization

INPUT/OUTPUT FORMAT

- **Transformer models:**

[CLS]	What is the capital of France?	[SEP]	Paris	[SEP]	} 5x
[CLS]	What is the capital of France?	[SEP]	Rome	[SEP]	
[CLS]	What is the capital of France?	[SEP]	Berlin	[SEP]	
...					

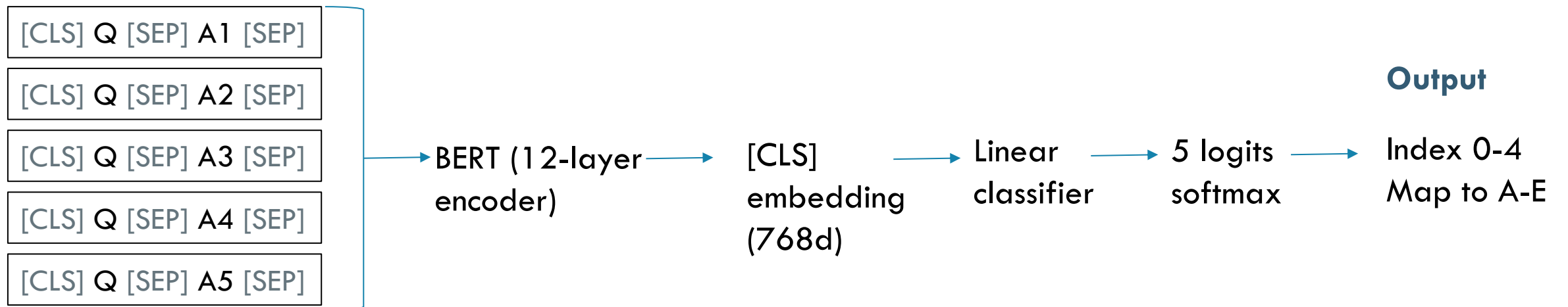
- **LLM:**

```
Question: What is the capital of France?  
A. Paris  
B. Rome  
C. Berlin  
D. London  
E. Madrid  
Answer:
```

Output

→ A

MODEL ARCHITECTURES: TRANSFORMER



MODEL ARCHITECTURE: FLAN-T5-XL

- Text-to-text
- 2.85B params
- open source
- trained on 1.8k+ tasks



EXPERIMENTS TRANSFORMER MODELS

- Grid search over:
 - Learning Rate: $1e-3$, $1e-4$, $1e-5$
 - Weight Decay: 0.01, 0.1
 - Scheduler Type: linear, cosine
- 12 runs per model
- CrossEntropyLoss & AdamW optimizer
- Batch size: 64
- Early Stopping (patience 12 vs. 8)
- Epochs: 30 vs. 20
- Best model by validation accuracy

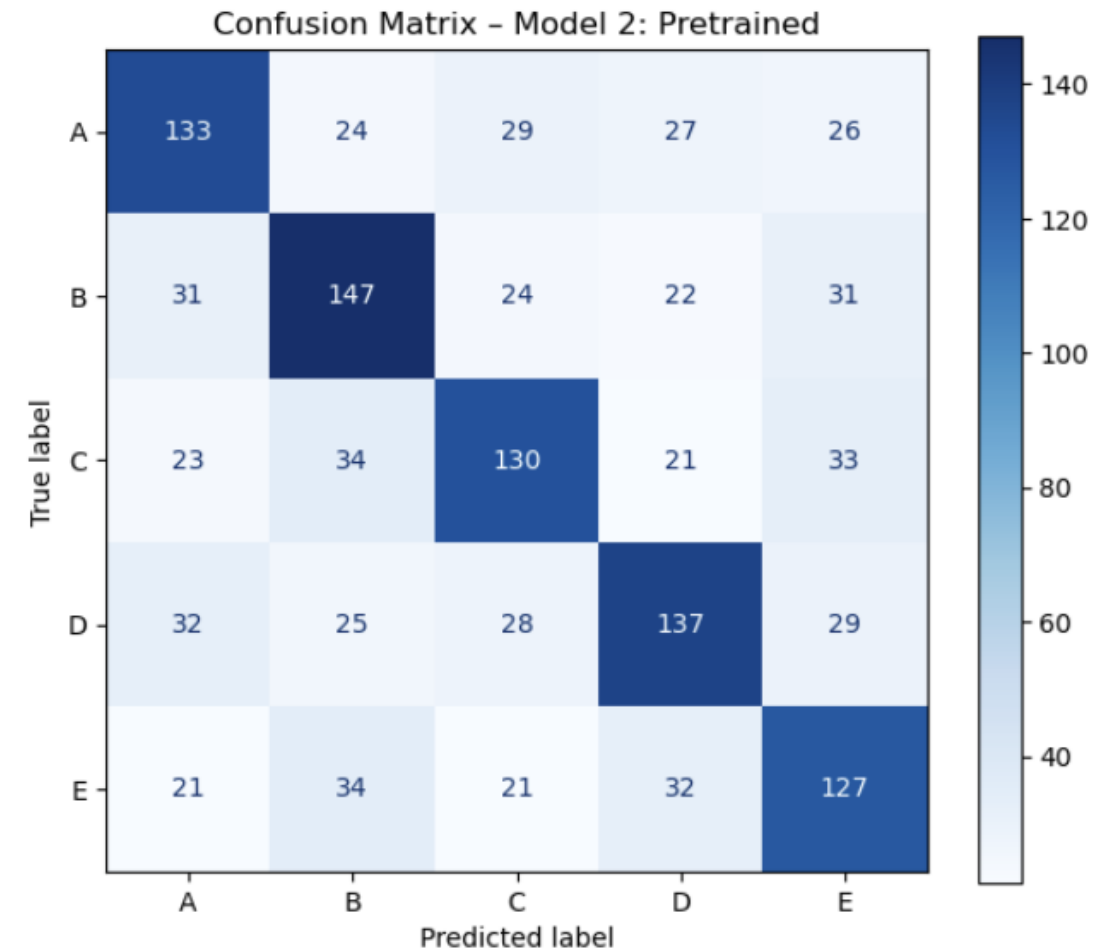
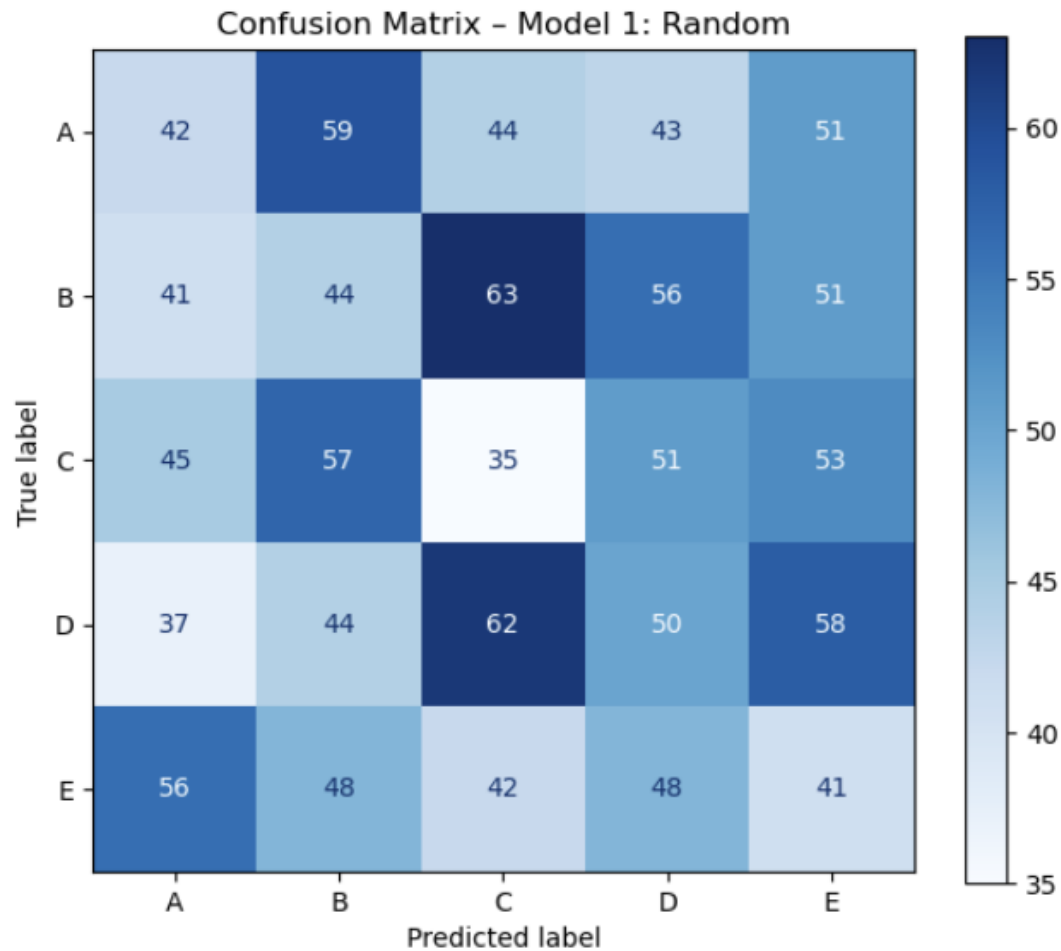
Best Setup

Parameter	Random	Pretrained
Learning Rate	$1e-4$	$1e-3$
Weight Decay	0.01	cosine
Scheduler	0.01	linear

EXPERIMENTS LLM: FLAN-T5-XL

- Zero-Shot vs. Few Shot
- same prompt format
- 3 examples from dataset for few-shot
- compared both setups on same test split

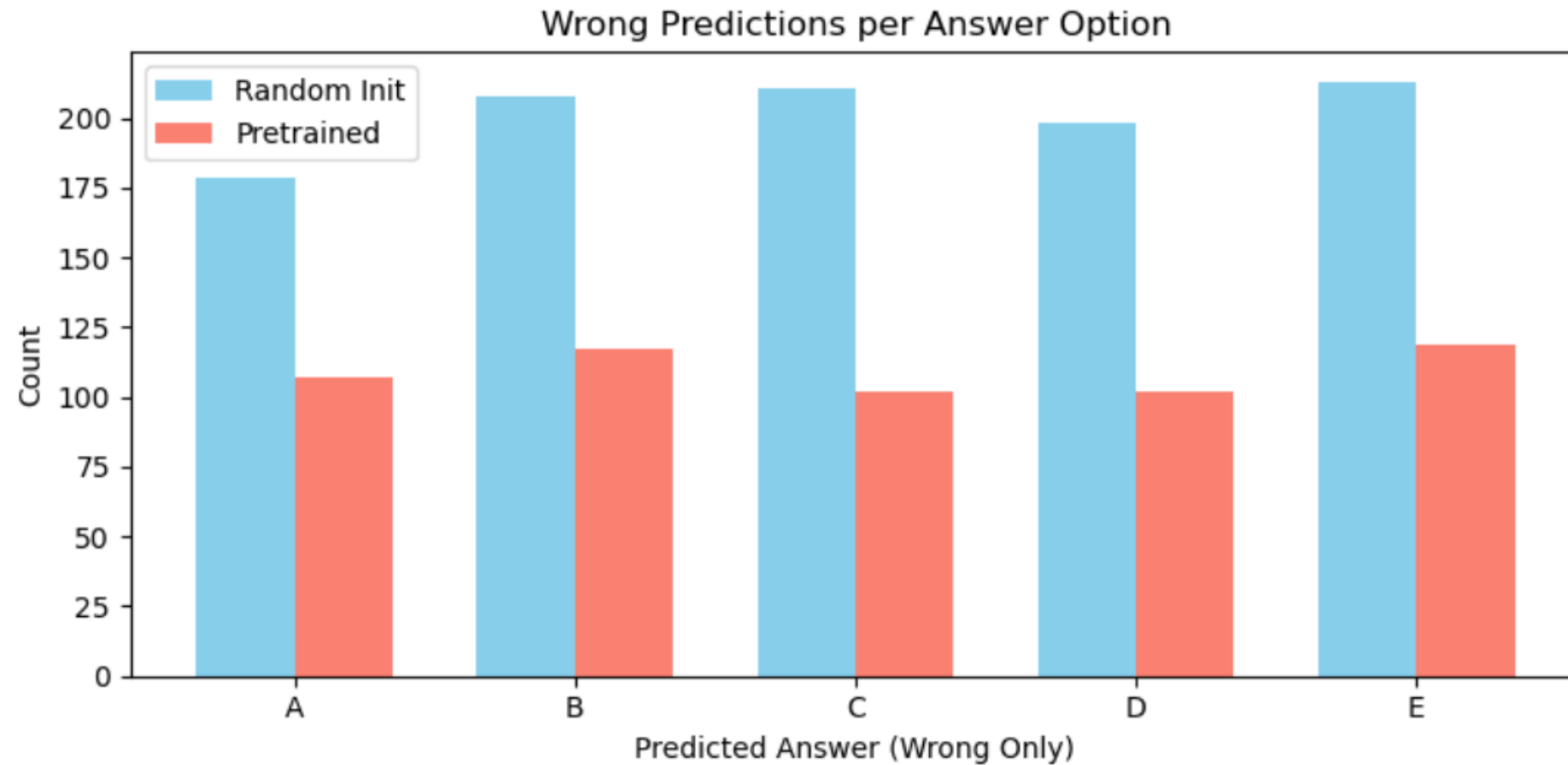
RESULTS



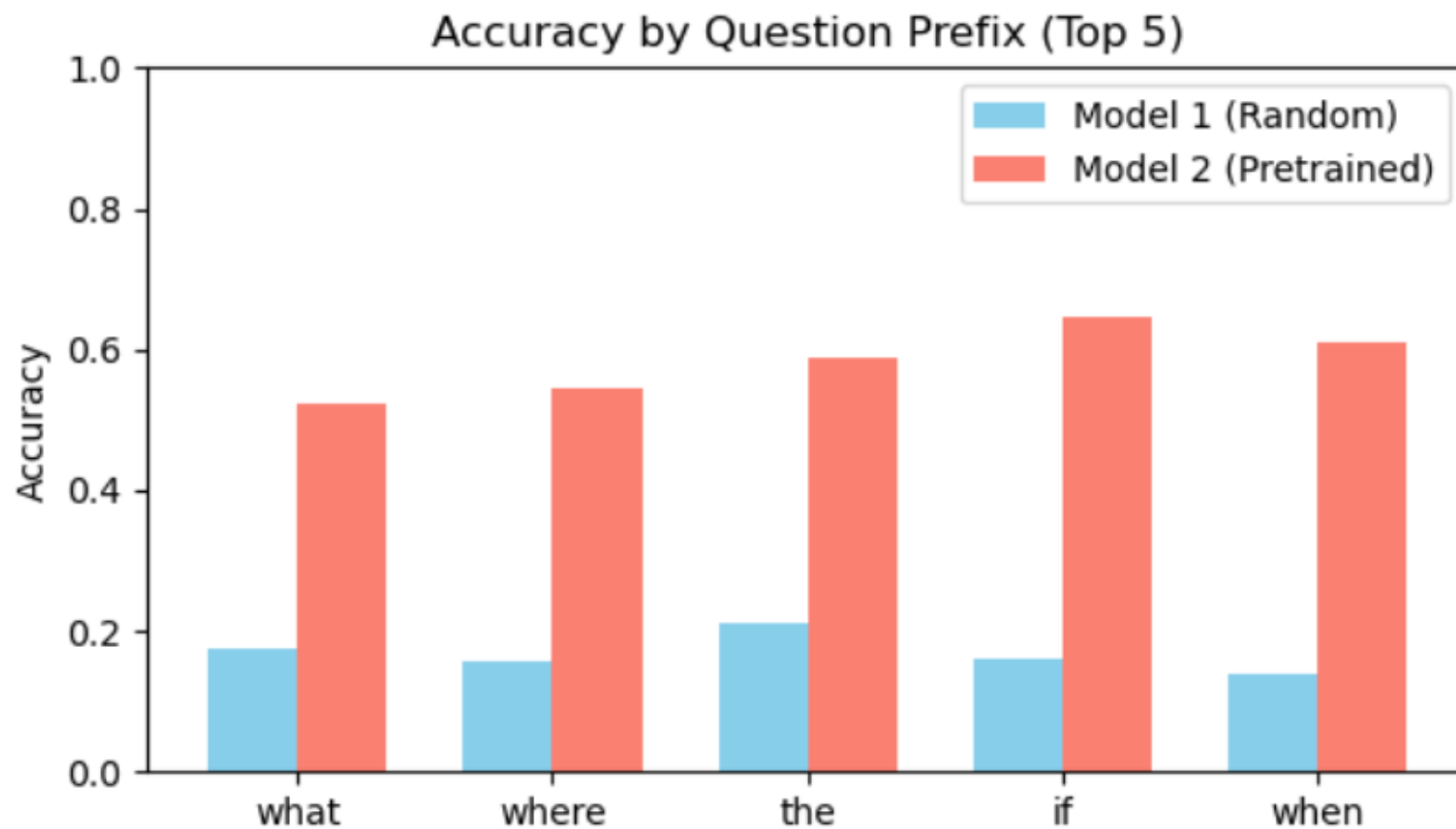
RESULTS

Model	Validation Accuracy	Test Accuracy
Baseline	20.0 %	20.0 %
Random BERT	22.1 %	17.4 %
Pretrained BERT	57.2 %	55.2 %
FLAN-T5-XL (zero-shot)	–	88.7 %
FLAN-T5-XL (few-shot)	–	89.2 %

RESULTS

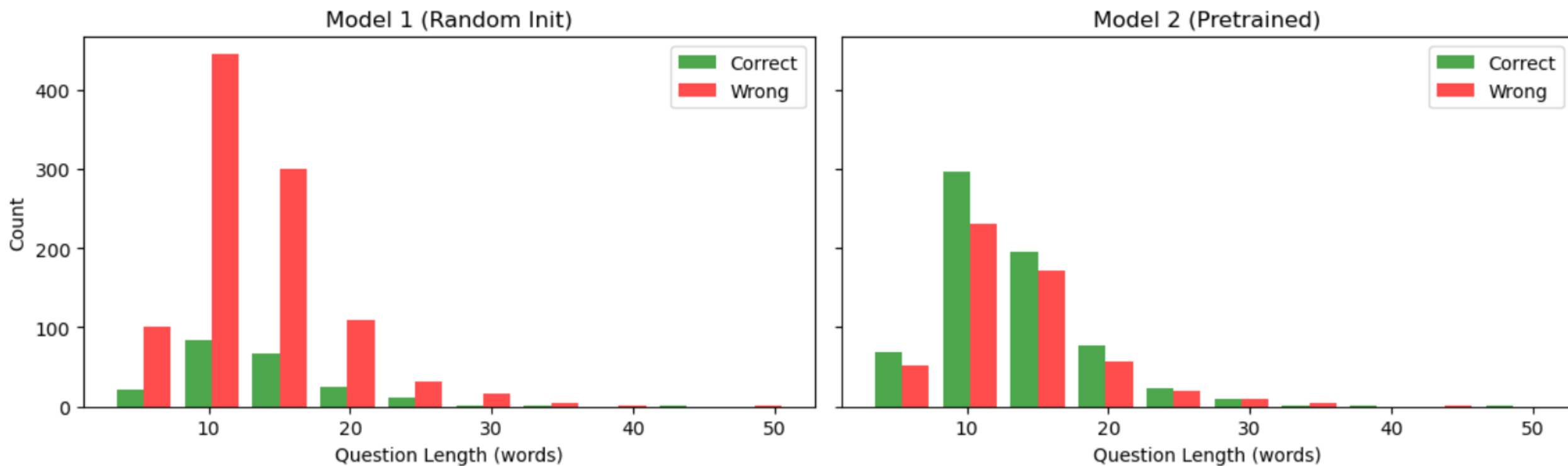


RESULTS



RESULTS

Question Length Distribution by Correctness

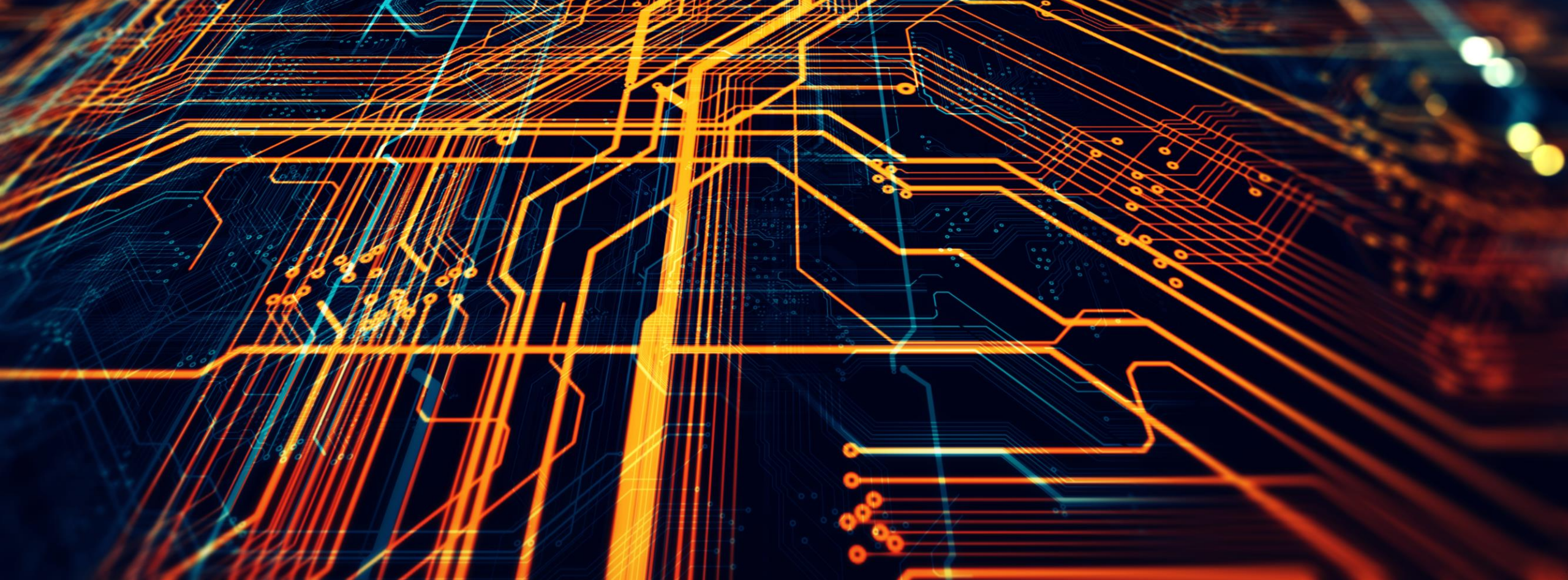


ERROR ANALYSIS

- Printed 10 misclassified examples/model
 - *random selection*
 - *If you want harmony, what should you do? -> make peace vs. make war*
 - *pretrained more reasonable*
- 10 misclassified examples for LLM
 - *Where might an unused chess set be stored? -> cupboard vs. attic*
- Questions where both BERT models incorrect but LLM correct
 - *379 in total*

INTERPRETATION

- Pretraining not optional
- Data & training matter more than architecture alone
- Prompting can outperform fine-tuning when used strong LLMs



THANK YOU FOR YOUR ATTENTION!